

Computing Theory and Practices (2026)

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M1U2

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M1U2 Plan

- 1 Complex vector spaces
- 2 Bra ket notation
- 3 Hermitian operators
- 4 Unitary operators
- 5 Expectation values
- 6 Unitary transformations
- 7 Position, Momentum, and Kinetic energy operators
- 8 Time dependent Schrodinger equation
- 9 Trapped atoms and ions
- 10 Superconductivity and Meissner effect
- 11 Cooper pairs and BCS theory
- 12 Josephson junctions and superconducting qubits

After completing this lecture, you will be able to

- Define a complex vector space.
- Distinguish between real and complex vector spaces.
- Perform vector addition and complex scalar multiplication.
- Explain linear combinations and basis vectors.
- Describe the role of complex vector spaces in quantum computing.

From Real to Complex Vector Spaces

A vector space consists of

- vectors,
- vector addition,
- scalar multiplication.

Examples of real vector spaces

$$\mathbb{R}^2, \quad \mathbb{R}^3$$

In quantum mechanics, the scalars are complex numbers.

$$\boxed{\mathbb{C}^2}$$

is the state space of a single qubit.

What is a Complex Vector Space?

A complex vector space is a vector space whose scalars belong to

$$\mathbb{C}.$$

Example

$$\mathbf{v} = \begin{bmatrix} 1 + i \\ 2 - i \end{bmatrix}$$

where

$$i = \sqrt{-1}.$$

Both the vector components and the multiplying scalars may be complex.

Operations in Complex Vector Spaces

Consider

$$\mathbf{u} = \begin{bmatrix} 1+i \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ -i \end{bmatrix}$$

Vector Addition

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} 4+i \\ 2-i \end{bmatrix}$$

Complex Scalar Multiplication

$$(2+i) \begin{bmatrix} 1 \\ i \end{bmatrix} = \begin{bmatrix} 2+i \\ -1+2i \end{bmatrix}$$

Linear Combinations and Basis

A linear combination is

$$\mathbf{w} = a\mathbf{u} + b\mathbf{v},$$

where

$$a, b \in \mathbb{C}.$$

For a single qubit,

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

form the computational basis.

Every qubit is a linear combination of these basis vectors.

Complex Inner Product

For vectors

$$|u\rangle \quad \text{and} \quad |v\rangle,$$

the inner product is

$$\langle u|v\rangle = u^\dagger v.$$

Example

$$|u\rangle = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

then

$$\langle u| = \begin{bmatrix} 1 & -i \end{bmatrix}.$$

The complex conjugate ensures physically meaningful probabilities.

Why Complex Numbers in Quantum Mechanics?

Complex numbers are essential because they describe

- Phase
- Interference
- Quantum superposition
- Time evolution

Euler's Formula

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Quantum amplitudes carry both magnitude and phase.

Worked Example

Given

$$\mathbf{u} = \begin{bmatrix} 1 + i \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ -i \end{bmatrix}$$

Compute

- 1 $\mathbf{u} + \mathbf{v}$
- 2 $i\mathbf{u}$
- 3 $\mathbf{u}^\dagger$

Discuss the role of complex conjugation while obtaining

$$\mathbf{u}^\dagger = [1 - i \quad 2].$$

Summary and Concept Check

Summary

- Complex vector spaces generalize real vector spaces.
- Scalars and vector components may be complex.
- Qubit states belong to $\mathbb{C}^2$.
- Linear combinations describe quantum superposition.
- Complex inner products define orthogonality and normalization.
- Complex numbers encode quantum phase.

Concept Check

- 1 What is the difference between $\mathbb{R}^2$ and $\mathbb{C}^2$?
- 2 Why is the complex conjugate required in the inner product?
- 3 Why are complex numbers indispensable in quantum mechanics?
- 4 How is a qubit represented in a complex vector space?

Learning Outcomes

After this lecture, you will be able to

- Explain the purpose of Dirac notation.
- Represent quantum states using ket vectors.
- Construct bra vectors from ket vectors.
- Calculate inner and outer products.
- Apply bra-ket notation to simple quantum systems.

Why Do We Need Dirac Notation?

Quantum states can be written as ordinary column vectors.

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Paul Dirac introduced a compact notation:

$$|0\rangle, \quad |1\rangle$$

Advantages

- Compact mathematical notation
- Easier representation of quantum states
- Simplifies quantum computations
- Standard notation in quantum mechanics

Ket Vectors

A **ket** represents a quantum state as a column vector.

Computational basis states

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

General qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where

$$|\alpha|^2 + |\beta|^2 = 1.$$

Bra Vectors

A **bra** is obtained by taking the conjugate transpose of a ket.

If

$$|\psi\rangle = \begin{bmatrix} 1 + i \\ 2 \end{bmatrix},$$

then

$$\langle\psi| = [1 - i \quad 2]$$

In general,

$$\langle\psi| = (|\psi\rangle)^\dagger$$

where $\dagger$ denotes the Hermitian (conjugate transpose) operation.

Inner Product (Bra–Ket)

The inner product is written as

$$\langle\phi|\psi\rangle.$$

It produces a scalar.

Examples

$$\langle 0|0\rangle = 1$$

$$\langle 1|1\rangle = 1$$

$$\langle 0|1\rangle = 0$$

The inner product measures

- overlap between states,
- orthogonality,

Outer Product (Ket–Bra)

The outer product is

$$|\psi\rangle\langle\psi|.$$

Example

$$|0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Unlike the inner product,
the outer product produces a matrix.

Applications

- Density matrices
- Projection operators
- Quantum measurements

Bra-Ket Operations

Expression	Result
$\langle\psi \phi\rangle$	Scalar
$ \psi\rangle\langle\phi $	Matrix
$A \psi\rangle$	New ket
$\langle\psi A$	New bra
$\langle\psi A \psi\rangle$	Expectation value

Always verify that matrix dimensions are compatible before performing operations.

Worked Example

Given

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

find

$$\langle\psi|, \quad \langle\psi|\psi\rangle, \quad |\psi\rangle\langle\psi|.$$

Solution

$$\langle\psi| = \frac{1}{\sqrt{2}} [1 \quad 1]$$

$$\langle\psi|\psi\rangle = 1$$

$$|\psi\rangle\langle\psi| = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Summary and Concept Check

Summary

- Ket vectors represent quantum states.
- Bra vectors are conjugate transposes of kets.
- Inner products produce scalars.
- Outer products produce matrices.
- Dirac notation is the standard language of quantum mechanics and quantum computing.

Concept Check

- 1 What is the difference between a bra and a ket?
- 2 Why is the complex conjugate required when forming a bra?
- 3 What is the difference between an inner product and an outer product?
- 4 Name two applications of the outer product in quantum computing.

Learning Outcomes

After this lecture, you will be able to

- Define a Hermitian operator.
- Test whether a matrix is Hermitian.
- Explain the physical significance of Hermitian operators.
- Identify the properties of Hermitian operators.
- Relate Hermitian operators to quantum measurements.

What is a Hermitian Operator?

A Hermitian operator is an operator that is equal to its own conjugate transpose.

Mathematically,

$$A^\dagger = A$$

where

$$A^\dagger$$

denotes the Hermitian (conjugate transpose) of A .

Hermitian operators represent measurable physical quantities in quantum mechanics.

Examples of Hermitian Matrices

Example 1

$$A = \begin{bmatrix} 2 & i \\ -i & 3 \end{bmatrix}$$

Since

$$A^\dagger = A,$$

the matrix is Hermitian.

Example 2

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Since

$$B^\dagger \neq B,$$

the matrix is not Hermitian

Properties of Hermitian Operators

Hermitian operators possess several important mathematical properties.

- All eigenvalues are real.
- Eigenvectors corresponding to distinct eigenvalues are orthogonal.
- Expectation values are real.
- Hermitian matrices are always square matrices.

These properties make Hermitian operators suitable for describing physical observables.

Hermitian Operators as Observables

Every measurable quantity in quantum mechanics is represented by a Hermitian operator.

Examples

Observable	Operator
Position	$\hat{x}$
Momentum	$\hat{p}$
Energy (Hamiltonian)	$\hat{H}$
Spin	$\hat{S}$

Measurement outcomes correspond to the eigenvalues of these operators.

Expectation Value

The expectation value of an observable A is

$$\langle A \rangle = \langle \psi | A | \psi \rangle$$

Interpretation

- Average value obtained from many identical measurements.
- Always a real number if A is Hermitian.

Expectation values connect quantum states with measurable physical quantities.

Worked Example

Determine whether

$$A = \begin{bmatrix} 4 & 2+i \\ 2-i & 1 \end{bmatrix}$$

is Hermitian.

Solution

Take the conjugate transpose,

$$A^\dagger = \begin{bmatrix} 4 & 2+i \\ 2-i & 1 \end{bmatrix} = A.$$

Therefore,

A is Hermitian.

Applications in Quantum Computing

Hermitian operators play an important role in quantum computing.

Applications include

- Quantum measurements
- Hamiltonian simulation
- Quantum chemistry
- Variational Quantum Eigensolver (VQE)
- Quantum machine learning

Many quantum algorithms are designed to estimate the eigenvalues of Hermitian operators.

Summary and Concept Check

Summary

- A Hermitian operator satisfies $A^\dagger = A$.
- Hermitian operators represent physical observables.
- Eigenvalues of Hermitian operators are always real.
- Expectation values are real.
- Quantum measurements are associated with Hermitian operators.

Concept Check

- 1 What condition must a Hermitian operator satisfy?
- 2 Why are Hermitian operators used to represent observables?
- 3 Why must the eigenvalues be real?
- 4 Give two examples of Hermitian operators in quantum mechanics.

Learning Outcomes

After completing this lecture, you will be able to

- Define a unitary operator.
- Verify whether a matrix is unitary.
- Explain why quantum gates are unitary.
- Describe the properties of unitary operators.
- Relate unitary operators to quantum state evolution.

What is a Unitary Operator?

A unitary operator preserves the norm of a quantum state.

Mathematically,

$$U^\dagger U = UU^\dagger = I$$

where

- $U^\dagger$ is the conjugate transpose of U ,
- I is the identity matrix.

Unitary operators describe the evolution of isolated quantum systems.

Properties of Unitary Operators

Important properties include

- Preserve vector length (normalization).
- Preserve inner products.
- Preserve probabilities.
- Are reversible.
- Their inverse is

$$U^{-1} = U^{\dagger}.$$

These properties ensure that quantum evolution is physically valid.

Example of a Unitary Matrix

Consider the Pauli-X gate

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Since

$$X^\dagger = X,$$

and

$$X^\dagger X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I,$$

the Pauli-X matrix is unitary.

Quantum Gates are Unitary

Every quantum gate is represented by a unitary matrix.

Examples

Identity Gate	I
Pauli-X Gate	X
Pauli-Y Gate	Y
Pauli-Z Gate	Z
Hadamard Gate	H

Quantum computation is a sequence of unitary transformations.

Quantum State Evolution

If the initial quantum state is

$$|\psi\rangle,$$

then after applying a quantum gate,

$$|\psi'\rangle = U|\psi\rangle$$

Since U is unitary,

$$\langle\psi'|\psi'\rangle = 1.$$

Normalization and total probability remain unchanged.

Worked Example

Apply the Pauli-X gate to

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Solution

$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle.$$

Thus, the Pauli-X gate acts as a quantum NOT gate.

Applications in Quantum Computing

Unitary operators are fundamental to

- Quantum gates
- Quantum algorithms
- Quantum simulation
- Quantum Fourier Transform
- Quantum teleportation
- Quantum error correction

Every computation on a quantum computer is implemented through a sequence of unitary operations.

Summary and Concept Check

Summary

- A unitary operator satisfies

$$U^\dagger U = I.$$

- Unitary operators preserve normalization.
- Quantum gates are unitary matrices.
- Quantum state evolution is described by

$$|\psi'\rangle = U|\psi\rangle.$$

- Unitary evolution is reversible.

Concept Check

- 1 What condition defines a unitary operator?
- 2 Why must quantum gates be unitary?
- 3 Why is reversibility important in quantum computation?
- 4 Give two examples of unitary matrices.

Learning Outcomes

After completing this lecture, you will be able to

- Define the expectation value of a quantum observable.
- Explain its physical interpretation.
- Calculate expectation values using bra-ket notation.
- Calculate expectation values using density matrices.
- Relate expectation values to quantum measurements.

Why Do We Need Expectation Values?

A single quantum measurement gives one of the possible eigenvalues of an observable.

Repeating the same measurement on many identically prepared systems produces a distribution of outcomes.

The average of these outcomes is called the **expectation value**.

Key Idea

The expectation value predicts the average result of many identical measurements, not the outcome of a single measurement.

Mathematical Definition

For a normalized quantum state $|\psi\rangle$ and a Hermitian operator A ,

$$\langle A \rangle = \langle \psi | A | \psi \rangle$$

where

- $|\psi\rangle$ is the state vector.
- A represents a measurable observable.
- $\langle A \rangle$ is the expectation value.

Since A is Hermitian, the expectation value is always real.

Physical Interpretation

Consider measuring the spin of an electron along the z-axis.

Possible measurement outcomes are

$$+1 \quad \text{or} \quad -1.$$

If the experiment is repeated many times, the average outcome approaches the expectation value.

Important

An expectation value may be a number such as 0.6, even though no individual measurement ever yields 0.6.

Worked Example Using Bra-Ket Notation

Given

$$|\psi\rangle = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

and the Pauli-Z operator

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},$$

find the expectation value.

Solution

$$\langle Z \rangle = \langle 0|Z|0\rangle = 1.$$

Hence, repeated measurements always produce the value +1.

Expectation Values Using Density Matrices

For a density matrix ρ , the expectation value is

$$\langle A \rangle = \text{Tr}(\rho A)$$

This expression is valid for

- Pure states
- Mixed states

The trace formulation is widely used in quantum computing and quantum information theory.

Worked Example Using Density Matrix

Consider

$$\rho = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

and

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Then

$$\langle Z \rangle = \text{Tr}(\rho Z) = 0.$$

Interpretation:

The probabilities of obtaining $+1$ and -1 are equal.

Applications of Expectation Values

Expectation values play a central role in modern quantum technologies.

Applications include

- Quantum measurements
- Variational Quantum Eigensolver (VQE)
- Quantum chemistry
- Quantum machine learning
- Quantum sensing
- IBM Qiskit Estimator Primitive

Many quantum algorithms estimate expectation values instead of reconstructing the complete quantum state.

Summary and Concept Check

Summary

- The expectation value is the average outcome of repeated measurements.
- For state vectors,

$$\langle A \rangle = \langle \psi | A | \psi \rangle.$$

- For density matrices,

$$\langle A \rangle = \text{Tr}(\rho A).$$

- Hermitian operators represent measurable observables.
- Expectation values are fundamental to quantum algorithms and experiments.

Concept Check

- 1 Why is the expectation value not necessarily an individual measurement outcome?
- 2 Why must observables be represented by Hermitian operators?

Learning Outcomes

After completing this lecture, you will be able to

- Explain the concept of a unitary transformation.
- Describe how unitary matrices transform quantum states.
- Verify that unitary transformations preserve normalization.
- Interpret quantum gates as unitary transformations.
- Apply simple unitary transformations to qubits.

What is a Unitary Transformation?

A unitary transformation changes a quantum state without changing its total probability.

If

$$|\psi\rangle$$

is the initial state, then the transformed state is

$$|\psi'\rangle = U|\psi\rangle$$

where

$$U^\dagger U = I.$$

Unitary transformations describe the evolution of isolated quantum systems.

Properties of Unitary Transformations

A unitary transformation has the following properties.

- Preserves normalization.
- Preserves inner products.
- Preserves probabilities.
- Is reversible.
- Conserves quantum information.

These properties ensure that quantum evolution is physically meaningful.

Transformation of a Quantum State

Consider the initial state

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Applying the Pauli-X gate,

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

gives

$$X|0\rangle = |1\rangle.$$

The unitary transformation changes the state while preserving normalization.

Hadamard Transformation

The Hadamard gate is

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Applying it to

$$|0\rangle$$

produces

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

A unitary transformation can create quantum superposition.

Preservation of Normalization

Suppose

$$|\psi'\rangle = U|\psi\rangle.$$

Then

$$\langle\psi'|\psi'\rangle = \langle\psi|U^\dagger U|\psi\rangle.$$

Since

$$U^\dagger U = I,$$

we obtain

$$\boxed{\langle\psi'|\psi'\rangle = \langle\psi|\psi\rangle = 1.}$$

Thus, normalization is preserved.

Unitary Transformations in Quantum Circuits

Quantum circuits are sequences of unitary transformations.

Examples

- Pauli-X gate
- Hadamard gate
- Phase gate
- Controlled-NOT (CNOT) gate

Each gate transforms the quantum state while preserving quantum information.

Worked Example

Given

$$|\psi\rangle = |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

apply the Pauli-X gate.

Solution

$$X|1\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle.$$

The Pauli-X gate swaps the computational basis states.

Summary and Concept Check

Summary

- A unitary transformation maps one quantum state to another.
- Quantum state evolution is described by

$$|\psi'\rangle = U|\psi\rangle.$$

- Unitary transformations preserve normalization and probabilities.
- Quantum gates implement unitary transformations.
- Quantum circuits are sequences of unitary transformations.

Concept Check

- 1 What is the difference between a unitary operator and a unitary transformation?
- 2 Why must normalization be preserved?
- 3 Which quantum gate creates superposition?
- 4 Why are unitary transformations reversible?

Learning Outcomes

After completing this lecture, you will be able to

- Explain why physical quantities are represented by operators.
- Write the position operator.
- Write the momentum operator.
- Derive the kinetic energy operator.
- Compute simple expectation values using these operators.

Operators as Physical Observables

Every measurable physical quantity is represented by a Hermitian operator.

Examples

$$\hat{x} \quad \hat{p} \quad \hat{T}$$

where

- $\hat{x}$: Position operator
- $\hat{p}$: Momentum operator
- $\hat{T}$: Kinetic energy operator

Measurement outcomes correspond to the eigenvalues of these operators.

Position Operator

For a particle in one dimension,

$$\hat{x} = x$$

The operator simply multiplies the wavefunction.

If

$$\psi(x)$$

is the wavefunction,
then

$$\hat{x}\psi(x) = x\psi(x)$$

The position operator is Hermitian.

Momentum Operator

According to quantum mechanics,

$$\hat{p} = -i\hbar \frac{d}{dx}$$

Acting on a wavefunction,

$$\hat{p}\psi(x) = -i\hbar \frac{d\psi}{dx}.$$

The momentum operator is Hermitian and represents linear momentum.

Kinetic Energy Operator

Classically,

$$T = \frac{p^2}{2m}.$$

Replacing momentum by its operator gives

$$\hat{T} = \frac{\hat{p}^2}{2m}$$

Substituting

$$\hat{p} = -i\hbar \frac{d}{dx},$$

we obtain

$$\hat{T} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}.$$

Expectation Values

Expectation value of position

$$\langle x \rangle = \langle \psi | \hat{x} | \psi \rangle$$

Expectation value of momentum

$$\langle p \rangle = \langle \psi | \hat{p} | \psi \rangle$$

Expectation value of kinetic energy

$$\langle T \rangle = \langle \psi | \hat{T} | \psi \rangle$$

Worked Example

Consider

$$\psi(x) = Ae^{-ax}.$$

Find the action of the momentum operator.

Solution

$$\begin{aligned}\hat{p}\psi(x) &= -i\hbar \frac{d}{dx}(Ae^{-ax}) \\ &= i\hbar a Ae^{-ax}.\end{aligned}$$

The operator transforms one wavefunction into another.

Applications

These operators are used in

- Schrödinger equation
- Quantum tunneling
- Quantum harmonic oscillator
- Quantum chemistry
- Semiconductor physics
- Quantum simulations

Most quantum mechanical models are built using these fundamental operators.

Summary and Concept Check

Summary

- Physical observables are represented by Hermitian operators.
- Position operator

$$\hat{x} = x.$$

- Momentum operator

$$\hat{p} = -i\hbar \frac{d}{dx}.$$

- Kinetic energy operator

$$\hat{T} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}.$$

- Expectation values provide measurable averages.

Concept Check

1 Why is the momentum operator differential in nature?

2 How is the kinetic energy operator obtained?

Learning Outcomes

After completing this lecture, you will be able to

- State the Time-Dependent Schrödinger Equation.
- Explain the physical meaning of each term.
- Describe the time evolution of a quantum state.
- Identify the Hamiltonian operator.
- Explain why the Schrödinger equation is central to quantum computing.

Why Do We Need the Schrödinger Equation?

In classical mechanics,

- Newton's laws predict the motion of particles.

In quantum mechanics,

- the state of a particle is described by a wavefunction,
- we require an equation that predicts how this wavefunction changes with time.

Key Idea

The Time-Dependent Schrödinger Equation governs the evolution of every isolated quantum system.

The Time-Dependent Schrödinger Equation

The fundamental equation of quantum mechanics is

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = \hat{H} \psi(x, t).$$

where

- $\psi(x, t)$: wavefunction
- $i = \sqrt{-1}$
- $\hbar$: reduced Planck constant
- $\hat{H}$: Hamiltonian operator

Hamiltonian Operator

The Hamiltonian represents the total energy of the system.

For a particle moving in a potential,

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$$

where

- first term: kinetic energy
- second term: potential energy

Physical Interpretation

The Schrödinger equation tells us

- how the quantum state changes with time,
- how probability amplitudes evolve,
- how energy influences the evolution.

Unlike Newton's equations, the Schrödinger equation predicts the evolution of the *wavefunction*, not a definite trajectory.

Time Evolution of Quantum States

If the Hamiltonian does not depend on time, the solution can be written as

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\psi(0)\rangle.$$

The operator

$$U(t) = e^{-i\hat{H}t/\hbar}$$

is called the **time evolution operator**.

It is a unitary operator.

Worked Example

Consider a free particle,

$$V(x) = 0.$$

The Hamiltonian becomes

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}.$$

The Time-Dependent Schrödinger Equation reduces to

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2}.$$

This equation describes the propagation of a free-particle wavefunction.

The Time-Dependent Schrödinger Equation is fundamental to

- Quantum computing
- Quantum tunneling
- Quantum chemistry
- Semiconductor devices
- Quantum optics
- Quantum simulations

Every quantum algorithm ultimately relies on the unitary time evolution described by this equation.

Summary and Concept Check

Summary

- The Schrödinger equation governs quantum dynamics.
- The Hamiltonian represents total energy.
- Quantum states evolve according to

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi.$$

- Time evolution is unitary.
- The equation forms the foundation of quantum mechanics and quantum computing.

Concept Check

- 1 What physical quantity does the Hamiltonian represent?
- 2 Why is the Schrödinger equation called the fundamental equation of quantum mechanics?
- 3 What is the role of the time evolution operator?
- 4 Why is quantum evolution unitary?

Learning Outcomes

After completing this lecture, you will be able to

- Explain the concept of trapped atoms and ions.
- Describe how ions are confined using electromagnetic fields.
- Explain how qubits are encoded in trapped ions.
- Identify the advantages and limitations of trapped-ion quantum computers.
- Compare trapped ions with other qubit technologies.

Why Trap Atoms and Ions?

Quantum computers require qubits that remain isolated from the environment.

Neutral atoms and ions are excellent candidates because they

- have well-defined energy levels,
- exhibit long coherence times,
- can be manipulated using lasers and electromagnetic fields.

The goal is to hold these particles nearly motionless so that quantum operations can be performed with high precision.

What is an Ion Trap?

An ion trap confines charged particles using electric and magnetic fields.

Two common types are

- Paul Trap (radio-frequency electric fields)
- Penning Trap (magnetic field + static electric field)

Ions are suspended in free space, avoiding contact with solid materials that could introduce decoherence.

Encoding a Qubit

A trapped ion has many electronic energy levels.

Two stable levels are selected to represent

$$|0\rangle \quad \text{and} \quad |1\rangle.$$

Laser pulses are used to

- initialize the qubit,
- perform quantum gates,
- read out the final state.

Quantum Operations

Laser pulses drive transitions between the qubit states.

Typical operations include

- State initialization
- Single-qubit rotations
- Two-qubit entangling gates
- State measurement using fluorescence

High-precision laser control enables very accurate quantum operations.

Advantages and Challenges

Advantages

- Long coherence time
- High gate fidelity
- Accurate measurements
- Well-understood technology

Challenges

- Slow gate speed
- Complex laser systems
- Difficult scaling
- Expensive experimental setup

Comparison with Other Qubit Platforms

Platform	Coherence	Gate Speed	Scalability
Trapped Ions	Excellent	Moderate	Moderate
Superconducting Qubits	Moderate	Excellent	High
Neutral Atoms	Excellent	Moderate	High

Each platform offers a different balance between coherence, speed, and scalability.

Trapped-ion quantum computers are widely used in

- Quantum algorithm research
- Quantum simulation
- Precision spectroscopy
- Atomic clocks
- Quantum networking
- Quantum error-correction experiments

They continue to play a major role in advancing quantum information science.

Summary and Concept Check

Summary

- Trapped ions are one of the leading physical implementations of qubits.
- Electromagnetic fields confine ions in free space.
- Laser pulses initialize, manipulate, and measure qubits.
- Trapped-ion systems provide excellent coherence and high-fidelity gates.
- Scalability remains an active area of research.

Concept Check

- 1 Why are ions trapped before performing quantum computations?
- 2 What physical states represent $|0\rangle$ and $|1\rangle$?
- 3 Name two advantages of trapped-ion quantum computers.
- 4 Why is scaling trapped-ion systems challenging?

Learning Outcomes

After completing this lecture, you will be able to

- Explain the phenomenon of superconductivity.
- Define the critical temperature.
- Describe the Meissner Effect.
- Distinguish superconductors from perfect conductors.
- Explain why superconductors are used in quantum computers.

What is Superconductivity?

Superconductivity is a state of matter in which a material exhibits

- Zero electrical resistance
- Perfect diamagnetism

This phenomenon occurs when the material is cooled below a characteristic temperature called the

$$T_c$$

known as the **critical temperature**.

Examples

- Mercury
- Lead
- Niobium
- Aluminium

Electrical Resistance in Superconductors

For ordinary conductors,
electrical resistance decreases gradually as temperature decreases.

For superconductors,
the resistance suddenly drops to

$$0$$

when

$$T < T_c.$$

This allows electric current to flow without energy loss.

Critical Temperature

The critical temperature is the temperature below which superconductivity appears.

Examples

Material	T_c (K)
Mercury	4.2
Lead	7.2
Niobium	9.3

Modern superconducting quantum processors operate at temperatures of only a few millikelvin.

The Meissner Effect

When a material becomes superconducting, it expels magnetic fields from its interior.

This phenomenon is called the

Meissner Effect.

Consequences

- Perfect diamagnetism
- Magnetic levitation
- Stable superconducting state

Superconductor vs Perfect Conductor

Superconductor

Zero resistance

Expels magnetic field

Shows Meissner Effect

Quantum phenomenon

Perfect Conductor

Zero resistance

Does not necessarily expel magnetic field

No Meissner Effect

Idealized classical concept

Applications in Quantum Computing

Superconductors are used to fabricate

- Josephson Junctions
- Superconducting Qubits
- Resonators
- Quantum Interconnects

Advantages

- Low energy loss
- Long coherence time
- Fast quantum gate operations

Worked Example

A niobium sample has

$$T_c = 9.3 \text{ K}.$$

Question

Will the sample be superconducting at

$$5 \text{ K?}$$

Answer

Since

$$5 < 9.3,$$

the sample is below its critical temperature.

Therefore,

it is in the superconducting state.

Summary and Concept Check

Summary

- Superconductors exhibit zero electrical resistance.
- Superconductivity appears below the critical temperature.
- The Meissner Effect expels magnetic fields.
- Superconductors differ from perfect conductors.
- Superconductivity forms the basis of superconducting quantum computers.

Concept Check

- 1 What is the critical temperature?
- 2 What is the Meissner Effect?
- 3 Why is a superconductor different from a perfect conductor?
- 4 Why are superconductors useful for quantum computing?

Learning Outcomes

After completing this lecture, you will be able to

- Explain the concept of Cooper pairs.
- Describe electron–phonon interaction.
- Define the superconducting energy gap.
- State the basic ideas of the BCS theory.
- Explain why superconductivity is essential for superconducting qubits.

Why Does Superconductivity Occur?

Ordinary electrical conduction is limited by

- collisions with lattice atoms,
- impurities,
- lattice vibrations (phonons).

Below the critical temperature,
electrons behave differently.

Instead of moving independently,
they form bound pairs called

Cooper Pairs.

Cooper Pairs

A Cooper pair consists of

- two electrons,
- opposite spins,
- opposite momenta.

Although electrons repel each other electrostatically, an indirect attraction is produced through lattice vibrations (phonons).

These paired electrons move through the crystal without electrical resistance.

Electron-Phonon Interaction

An electron moving through the crystal slightly distorts the positive ion lattice.

This distortion attracts another electron, producing an effective attractive interaction.

The sequence is

Electron $\rightarrow$ Lattice Distortion $\rightarrow$ Phonon $\rightarrow$ Second Electron

This indirect attraction leads to the formation of Cooper pairs.

Energy Gap

When electrons form Cooper pairs,
an energy gap appears around the Fermi level.

The superconducting energy gap is denoted by

$$\Delta$$

Consequences

- prevents low-energy scattering,
- increases stability of Cooper pairs,
- maintains zero electrical resistance.

BCS Theory was proposed in 1957 by

- John Bardeen
- Leon Cooper
- John Robert Schrieffer

Main ideas

- Superconductivity arises from Cooper pair formation.
- Electron–phonon interaction creates an attractive force.
- Cooper pairs condense into a single quantum state.
- The energy gap protects the superconducting state.

Importance in Quantum Computing

Superconducting quantum processors use materials such as

- Aluminium
- Niobium

Cooper pairs carry current through superconducting circuits with almost no energy loss.

This enables

- long coherence times,
- low-noise quantum circuits,
- Josephson junction operation.

Worked Example

A superconducting material is cooled below its critical temperature.

Question

Why does its electrical resistance become nearly zero?

Answer

- Electrons form Cooper pairs.
- An energy gap suppresses scattering.
- Cooper pairs move coherently through the crystal lattice.

Hence, electrical resistance disappears.

Summary and Concept Check

Summary

- Cooper pairs are pairs of electrons with opposite spin and momentum.
- Electron–phonon interaction produces an effective attraction.
- The superconducting energy gap stabilizes the superconducting state.
- BCS Theory explains conventional superconductivity.
- Cooper pairs are the foundation of superconducting quantum computers.

Concept Check

- 1 What is a Cooper pair?
- 2 How does a phonon help two electrons attract each other?
- 3 What is the role of the superconducting energy gap?
- 4 Why is BCS Theory important for superconducting qubits?

Learning Outcomes

After completing this lecture, you will be able to

- Explain the structure of a Josephson junction.
- Describe the Josephson effect.
- Understand how superconducting qubits are realized.
- Identify different types of superconducting qubits.
- Explain the advantages and challenges of superconducting quantum computers.

What is a Josephson Junction?

A Josephson junction consists of

Superconductor – Insulator – Superconductor

The insulating layer is extremely thin (typically 1–2 nm).

Cooper pairs can tunnel through the insulating barrier without electrical resistance.

This quantum tunneling phenomenon is called the **Josephson Effect**.

The Josephson Effect

Two important effects occur in a Josephson junction.

DC Josephson Effect

- A supercurrent flows across the junction even when no voltage is applied.

AC Josephson Effect

- When a constant voltage is applied, the supercurrent oscillates with time.

These effects arise from the coherent tunneling of Cooper pairs.

From Josephson Junction to Qubit

A Josephson junction behaves as a nonlinear circuit element.

Unlike ordinary electrical circuits,
it possesses discrete quantum energy levels.

Two energy levels are chosen to represent

$$|0\rangle \quad \text{and} \quad |1\rangle.$$

Thus, the Josephson junction becomes a superconducting qubit.

Types of Superconducting Qubits

Common superconducting qubits include

- Charge Qubit
- Flux Qubit
- Phase Qubit
- Transmon Qubit

The **Transmon** is widely used because it offers improved coherence and reduced sensitivity to electrical noise.

Quantum Operations

Microwave pulses are used to control superconducting qubits.

Typical operations include

- State initialization
- Single-qubit gates
- Two-qubit gates
- State readout using microwave resonators

Quantum information is encoded in the coherent evolution of the qubit states.

Advantages and Challenges

Advantages

- Fast quantum gates
- Microfabrication compatible
- Scalable integrated circuits
- Well suited for quantum processors

Challenges

- Shorter coherence time
- Cryogenic cooling required
- Sensitive to environmental noise
- Calibration complexity

Applications

Superconducting qubits are used in

- IBM Quantum systems
- Google Quantum AI processors
- Rigetti quantum computers
- Quantum chemistry
- Quantum optimization
- Quantum machine learning

Today, superconducting qubits are among the leading hardware platforms for gate-based quantum computing.

Summary and Concept Check

Summary

- A Josephson junction is formed by two superconductors separated by a thin insulating barrier.
- Cooper pairs tunnel through the barrier via the Josephson effect.
- Josephson junctions possess discrete quantum energy levels.
- Two energy levels form the logical qubit states.
- Superconducting qubits power many modern quantum computers.

Concept Check

- 1 What is the structure of a Josephson junction?
- 2 What is the Josephson effect?
- 3 Why are Josephson junctions suitable for qubit implementation?
- 4 Which superconducting qubit architecture is most widely used today?